

DATABASE laboratory work 1

International Airport Database

This laboratory project designs and implements a relational database architecture for an International Airport operational system. The database models core airport workflows including flight logistics, passenger administration, reservation processing, luggage tracking, and security compliance. The relational schema is normalized into 10 distinct entities to ensure domain data integrity and eliminate functional redundancies.

1. airports: Represents international airport hubs.
 - Attributes: **airport_id** , airport_name, country, state, city, created_at, updated_at.
2. airlines: Stores commercial airline carrier metadata.
 - Attributes: **airline_id** , airline_code, name, country, created_at, updated_at.
3. flights: Tracks flight schedules, route origins/destinations, and real-time operational status.
 - Attributes: **flight_id** , **airline_id**, **departure_airport_id**, **arrival_airport_id**, departing_gate, arriving_gate, scheduled_departure_time, scheduled_arrival_time, actual_departure_time, actual_arrival_time, created_at, updated_at.
4. passengers: Maintains personal identity profiles and travel documentation.
 - Attributes: **passenger_id** , first_name, last_name, gender, date_of_birth, country_of_citizenship, country_of_residence, passport_number, created_at, updated_at.
5. booking: Logs ticket reservations connecting passengers to assigned flights.
 - Attributes: **booking_id** , **passenger_id**, **flight_id**, status, booking_platform, ticket_price, created_at, updated_at.
6. booking_changes: Audit log maintaining historical modifications to existing reservations.
 - Attributes: **booking_change_id** , **booking_id**, change_details, created_at, updated_at.
7. boarding: Controls issued boarding passes and seat allocations.
 - Attributes: **boarding_pass_id** , **booking_id**, seat_number, boarding_time, created_at, updated_at.
8. baggage: Records registered checked luggage dimensions linked to bookings.
 - Attributes: **baggage_id** , **booking_id** , weight_kg, created_at, updated_at.
9. baggage_checks: Logs inspection results for passenger luggage items.
 - Attributes: **baggage_check_id** , **booking_id** , **passenger_id** , check_results, created_at, updated_at.
10. security_check: Records passenger screening protocol results prior to boarding.
 - Attributes: **security_check_id** , **passenger_id** , check_results, created_at, updated_at.

Connection:

Connections connect tables with foreignn and primary key attributes. To understand connections better we can divide them into 3 sections:

Flight connection:

airlines -> flights (1:N): One airline does many flights.

airports -> flights (1:N): One airport has many departures and arrivals

Booking:

passengers → booking (1:N): One passenger can book many tickets.

flights → booking (1:N): One flights contains a lot bookings.

booking → booking_changes (1:N): One booking may have many changes.

booking → boarding (1:1): One booking has exactly one boarding with one seat and time.

Baggage:

booking → baggage (1:N): In one booking may have many baggages.

booking + passengers → baggage_checks (1:N): Baggage checks connects passenger and its booking, and there might be more than one baggage checks.

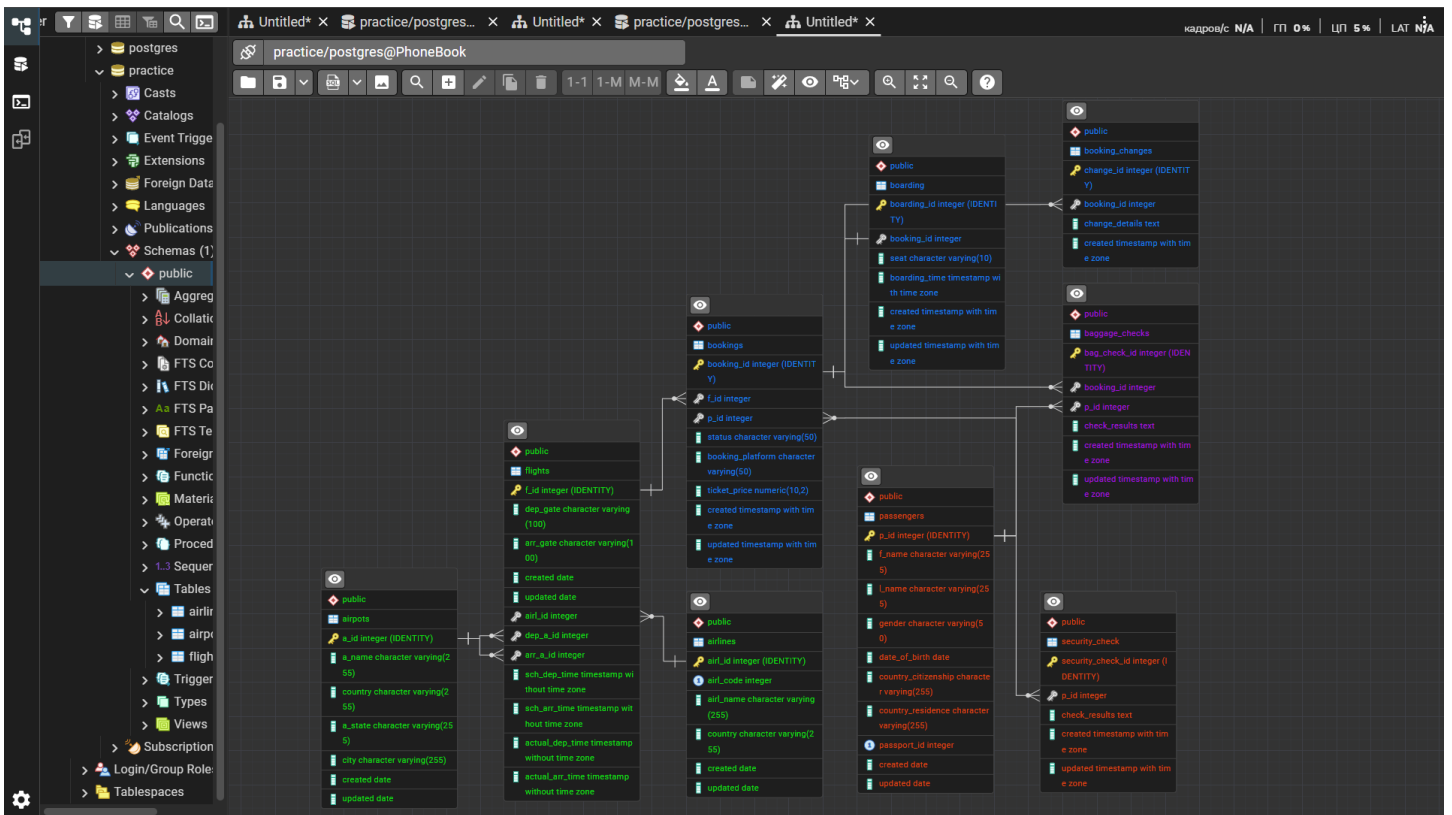
passengers → security_check (1:N): Passenger could went from many security checks.

Visualistion:

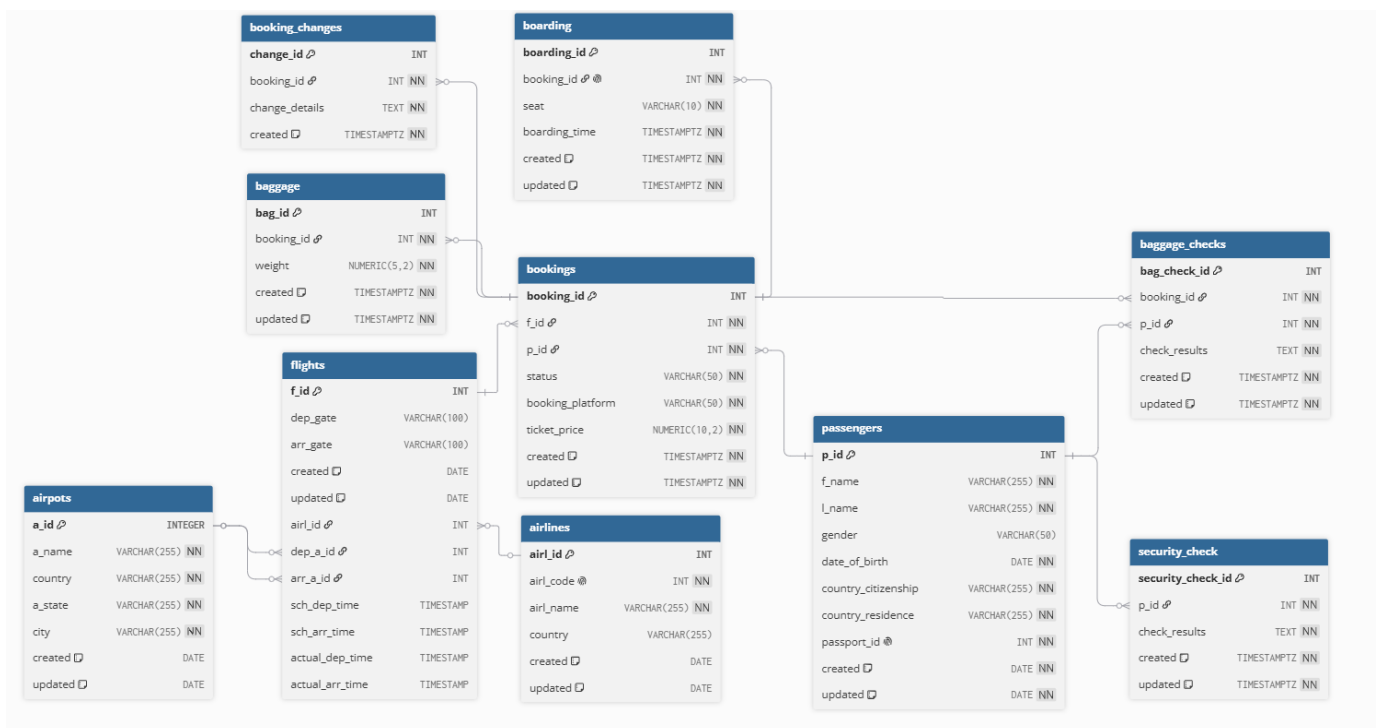
TABLE:

Name	Primary Key	Foreignn Key
Airports	airport	---
Airlines	airline id	---
Flights	flight id	airline id, departure time, arrival airport id
Passengers	passenger id	---
Booking	booking id	passenger id, flight id
Booking changes	booking changes id	passenger id
Boarding	boarding id	booking id
Baggage	baggage id	booking id
Baggage checks	baggage check id	booking id, passenger id
Security checks	security check id	passenger id

Picture 1: Physical Database ERD Generated via pgAdmin 4



Picture 2: Logical ERD Topology via dbdiagram.io



Picture 2: Chen Notation ERD

